

Education

Ohio University

M.S. Computer Science (4.0)
B.S. Electrical Engineering (3.9)
B.S. Computer Science (3.9)

Athens, OH
Expected August 2025
Completed May 2023
Completed May 2022

Work Experience

Graduate Researcher – Ohio University

Aug 2022 → Present

- Engineered a passive wireless temperature sensing system using patch antennas and Tensorflow-based LSTM models achieving a 1.4 °C mean absolute error in temperature estimates.
- Authoring a peer-reviewed publication for Q3 2025.

Adjunct Professor – Ohio University

Jan 2022 → Jan 2023

- Designed and taught CS4100 (Formal Languages and Compilers) to a class of 50 seniors; received positive student evaluations
- Created Rust-based assignments focused on formal languages and compilers, including custom lexer and recursive descent parser implementations for a subset of C-like syntax
- Conducted weekly lab sessions on embedded systems development using STM32 microcontrollers, covering GPIO, timers, and memory-mapped I/O in both C and ARM assembly

Software Engineering Intern – Rescon

Aug 2021 → Aug 2022

- Trained next-generation reservoir computing models using TensorFlow, integrated with PX4 on MODALAI drones for real-time obstacle avoidance in GPS-denied environments

Projects

Laser Light Show: Built a modular C++ real-time game engine on Teensy 4.1 for a laser console projecting vector graphics via galvo-controlled mirrors; integrated custom DAC/amp hardware, UART controllers, and touchscreen UI

Player Needed: Built a cross-platform mobile app using React Native and Firebase to connect users through local sports and study events; implemented real-time chat, group timelines, and event filtering in a modular, scalable codebase

Andromeda Networks: Architected a scalable multiplayer game server with a custom Lua-based backend featuring economy and moderation systems; hosted over 100 active players

Honors and Leadership

Senior Engineer of the Year: Awarded by the Russ College of Engineering

Student Expo First Place: Won for Laser Light Show senior design project; competed against the entire senior class of the Russ College Electrical Engineering Department

Tau Beta Pi – Treasurer: Oversaw finances, prepared reports, and coordinated chapter events

Technical Skills

Coding Languages: C/C++, C#, Python, JavaScript, HTML, CSS, Rust, Lua

DevOps & Tools: Docker, Kubernetes, Git, Apache, Nginx, .NET

Cloud Platforms: Azure, AWS, DigitalOcean

Databases: MySQL, PostgreSQL, Firebase

Modeling & Simulation: MATLAB, Simulink

General Skills: Agile (Scrum/Kanban), Systems Programming, Fullstack Development, Machine Learning, Large Language Models, Project Management, Graphic Design (Adobe Suite)